Mini-Project Report

TY B.Tech Artificial Intelligence and Machine Learning

Verifying ICU Arrhythmia Alarms from ECG and Pulse Signals: Cross-Attention Fusion Compared with Feature-Based and Deep Learning Baselines

Submitted by

Veerbhadra Mahant
Ved Jadhav
Sharvil Patil

Under the guidance of

Dr. Roshani Raut
Department of Information Technology

Pimpri Chinchwad College of Engineering, Pune
Pune, India

veerbhadra.mahant24@pccoepune.org | ved.jadhav24@pccoepune.org
sharvil.patil24@pccoepune.org | roshani.raut@pccoepune.org

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Abstract

False alarms from ICU bedside monitors wear down staff attention, and an alarm that gets ignored can be a real one. The project asks whether a second signal, the pulse waveform (a photoplethysmogram, PPG, or arterial pressure where PPG is missing), can verify an arrhythmia alarm that the ECG has triggered, and whether cross-attention is the right way to combine the two. Two open PhysioNet datasets are used: the 2015 Computing in Cardiology Challenge training set (750 records, of which 592 form the PPG cohort) and VTaC (4,542 usable ventricular tachycardia alarms). Each input is the 10 s of ECG and pulse that end at the alarm onset, filtered and z-normalised. The cross-attention network is compared with ECG-only, PPG-only and concatenation models of equal size, other neural variants, six classical learners on 266 engineered features, an attention CNN-GRU recipe and ensembles, all under record-wise cross-validation with an inner validation set. After a validation leak in the first version of the code was fixed, cross-attention reached F1 0.699 and AUC 0.841 on Challenge 2015, no better than concatenation (F1 0.693). The attention CNN-GRU recipe reached F1 0.828 on Challenge 2015 and F1 0.858 (accuracy 0.913, AUC 0.966) on the VTaC test split. Counterfactual tests show that the networks use what each signal looks like, not the timing between ECG and pulse. The main limitations are small data, no prospective validation and weak transfer from Challenge 2015 to VTaC. Larger pretraining corpora and leave-one-site-out testing are the sensible next steps.

1. Introduction

1.1 Background

A bedside monitor in an intensive care unit (ICU) watches the electrocardiogram (ECG) around the clock and raises an alarm whenever the rhythm looks dangerous. Many of those alarms are wrong. In one review of 5,386 critical ECG arrhythmia alarms from 447 adult patient records, 42.7% were false [1]. After a shift of false alarms, staff answer more slowly. The literature calls this alarm fatigue [2], [3], and the rare true alarm gets the same slow answer.

The causes are mostly mundane: ECG artefacts and low-voltage signals on a single channel [1]. A loose electrode or a patient who shifts in bed corrupts the ECG, and the ECG is the same signal anyone would use to check the alarm. Ventricular tachycardia (VT) causes the most trouble. VT alarms are among the most frequent life-threatening alarms, and the false ones have been the hardest to detect reliably [4].

1.2 Importance of AI, ML and deep learning

Rule-based alarm logic and hand-built features both lean on a beat detector, and detectors tend to fail on the noisy segments where help is needed most. A deep network can take the raw waveform instead. An attention layer can go one step further and learn which stretch of one signal to compare with which stretch of another [5]. Published attention-based networks report sensitivity above 90% on the 2015 Challenge data [6], and that is the argument we started from.

We wanted to test that argument fairly, and the measurements only partly support it. With a few hundred labelled records, engineered features beat the plain deep networks on both datasets (Sections 7 and 8).

1.3 Problem definition

One example is one alarm event. We take the 10 seconds of ECG and pulse signal that end at the moment the monitor raised the alarm (the pulse is PPG, or arterial blood pressure when no PPG was recorded). At 250 Hz that gives each signal 2,500 samples. The model returns a probability that the alarm is true, so the task is binary classification. The intended use is decision support: rank or suppress alarms that are very likely false while keeping sensitivity to real events high. It is a research prototype and not a medical device.

1.4 Objectives

(1) Prepare leak-free, decision-time windows from two open ICU alarm datasets and define record-wise evaluation splits.

(2) Implement a cross-attention fusion network and compare it with ECG-only, PPG-only and concatenation baselines of equal parameter count.

(3) Benchmark it against other neural architectures, classical models on engineered features, a stronger deep training recipe and ensembles, using the same splits.

(4) Test with perturbation experiments whether the fusion models rely on the timing between ECG and pulse, and measure robustness to noise, sensor loss and a change of dataset.

(5) Report every result with corrected statistics, analyse misclassifications and state the limitations.

1.5 Scope

Included: retrospective experiments on the PhysioNet/CinC Challenge 2015 training set and the VTaC database, an AF-screening side task on MIMIC PERform AF, noise-robustness tests on MIT-BIH Noise Stress Test recordings, and a comparison with an open PPG foundation model. Excluded: prospective or bedside testing, any regulatory claim, data from outside US ICUs, demographic or skin-tone analysis of optical signals, and the other alarm types in VTaC (the database only contains VT alarms). The 500-record Challenge 2015 test set was never released, so no result here uses it.

2. Literature Survey

The idea of using a second physiological signal to question an ECG alarm is not new. Aboukhalil and colleagues built a suppression algorithm from the shape and timing of the arterial blood pressure waveform and tested it on alarms from the MIMIC II database. It cut the share of false critical ECG alarms from 42.7% to 17.2% where both signals were available, but it also suppressed 9.4% of true VT alarms, which the authors flagged as the reason for further refinement [1].

The 2015 PhysioNet/Computing in Cardiology Challenge gave the problem a shared benchmark. The organisers released 750 five-minute ICU segments for training and kept 500 back for testing. Entries were scored with a penalty that weights a suppressed true alarm five times more heavily than an accepted false one. In all, 38 teams submitted 215 entries [7], [8]. Only the 750 training records are public, and they are the data most later work uses, ours included.

Deep learning entered through that dataset. Mousavi, Fotoohinasab and Afghah combined convolutional layers, an attention mechanism and LSTM units, trained in two steps (pre-training, then fine-tuning), and reported 93.88% sensitivity and 92.05% specificity using three signals, and 90.71% sensitivity, 88.30% specificity and an AUC of 89.51 for VT alarms with a single ECG lead [6]. Zhou and co-authors added a similarity loss from pair-wise comparisons of waveform segments to the usual classification loss and fed in learned embeddings from a rule-based method for each alarm type; they report better performance than the Challenge’s winning entries [9]. We re-implemented the first of these as the starting point for our strongest recipe, working from the published description, so our version is not the authors’ code.

Data have grown since. VTaC offers more than 5,000 VT alarm recordings from bedside monitors, each labelled by at least two independent annotators, along with benchmark results for conventional, deep, semi-supervised and generative methods [4]. For the pulse signal, a benchmark of 15 PPG beat detectors found that eight did well at rest in hospital and wearable data, while accuracy dropped in atrial fibrillation, in neonates and during exercise [10]. Pretraining has moved too: PaPaGei, an open PPG foundation model trained on about 57,000 hours of data, improved classification metrics by 6.3% and regression metrics by 2.9% in at least 14 of 20 tasks [11].

Table 1: Studies reviewed, with results as quoted from the sources. "Not stated" means the abstract gives no number.

Study	Data	Method	Reported result	Limitation or relevance
Aboukhalil et al., 2008 [1]	MIMIC II: 5,386 critical ECG alarms, 447 adult records	Suppression rules from ABP morphology and timing	False alarms 42.7% to 17.2%; 59.7% of false alarms suppressed; true alarms lost: 0% except VT (9.4%)	Needs ABP; some true VT alarms suppressed; prospective evaluation recommended
Clifford et al., 2015 and 2016 [7], [8]	Challenge 2015: 750 training and 500 hidden test segments, 5 min each	Open challenge; 5x penalty for suppressed true alarms	38 teams, 215 entries; best score not stated	Only the 750 training records are public; defines the data and the score used here
Mousavi et al., 2020 [6]	Challenge 2015	CNN, attention, LSTM; pre-training then fine-tuning	Sens. 93.88%, spec. 92.05% (3 signals); VT, one ECG lead: sens. 90.71%, spec. 88.30%, AUC 89.51	Protocol not reproduced, so not directly comparable with ours
Zhou et al., 2022 [9]	Challenge 2015	CNN with classification plus similarity loss; rule-based alarm-type embeddings	Better than the winning Challenge entries (numbers not stated)	Needs alarm-type knowledge; figures not stated
Lehman et al., 2023 [4]	VTaC: over 5,000 VT alarms, at least 2 annotators	Benchmark of conventional, deep, semi-supervised and generative methods	Dataset and benchmark paper; results in the paper	VT alarms only; no detailed clinical information
Pillai et al., 2025 [11]	20 tasks, 10 datasets; pretrained on about 57,000 h of PPG	Open PPG foundation model (PaPaGei)	+6.3% classification, +2.9% regression in at least 14 tasks	Listed tasks do not include ICU alarm verification, so transfer had to be tested
Charlton et al., 2022 [10]	8 datasets, 15 PPG beat detectors	Benchmark framework for beat detection	Eight detectors F1 of 90% or more at rest; poorer in AF, neonates, exercise	Source of the MIMIC PERform AF data used in the AF side task

The abstracts do not say enough to line these papers up against each other: splits, validation sets and parameter budgets are left out. A fair comparison would use the same splits, the same parameter budget, a validation set separate from the test fold, and strong non-deep baselines beside the networks. That is what we tried to build here, on two datasets. We also wanted to know whether attention between ECG and pulse does what its design assumes.

3. Dataset Used

Two labelled datasets carry the main experiments, and Table 2 lists every data source we used. We checked the source pages on 14 October 2026.

Table 2: Data sources used in the experiments.

Dataset	Samples	Features	Classes / target	Source and citation
PhysioNet/CinC Challenge 2015, PPG cohort	592 records (724 in the recovered cohort)	2 channels x 2,500 samples (raw); 266 engineered features	True alarm (223), false alarm (369)	[7], [12]; physionet.org/content/challenge-2015/1.0.0/
VTaC v1.1	4,542 events kept of 5,037	2 channels x 2,500 samples (raw); 266 engineered features	True VT alarm (28.0%), false	[4], [13]; physionet.org/content/vtac/
MIMIC PERform AF	35 subjects; 2,100 windows used	ECG and PPG, 125 Hz	AF (19 subjects), non-AF (16)	[10], [14]; zenodo.org/records/15906524
MIT-BIH Noise Stress Test	noise recordings	baseline wander, electrode motion, muscle artefact	not applicable (noise source)	[15]; physionet.org/content/nstdb/1.0.0/

3.1 PhysioNet/CinC Challenge 2015

The Challenge 2015 training set has 750 five-minute ICU records at 250 Hz. Each holds one bedside-monitor alarm, which fires at the 300 s mark, with an expert's true or false label for one of five alarm types [7], [8]. We read the recordings with the WFDB tools from PhysioNet [16]. Of the 750 records, 627 have a PPG channel, 343 have arterial blood pressure (ABP) and every record has a second ECG lead. The original PPG cohort keeps the 592 records with a usable PPG window, and 37.7% of them are true alarms. For a recovered cohort we used ABP as the pulse channel in 132 records that have no PPG and dropped 26 records whose window is too short or nearly flat. That cohort has 724 records, 39.6% true.

Table 3: Alarm types in the 592-record PPG cohort.

Alarm type	Records	True alarms	False alarms
Asystole	93	14	79
Extreme bradycardia	73	39	34
Extreme tachycardia	111	106	5
Ventricular flutter / fibrillation	45	5	40
Ventricular tachycardia	270	59	211
Total	**592**	**223**	**369**

The table shows how uneven the data are. Almost every tachycardia alarm is true, almost every flutter/fibrillation alarm is false, and asystole and flutter/fibrillation have too few true examples for per-type conclusions. VT, the largest group, is mixed.

3.2 VTaC

VTaC contains 5,037 annotated VT alarm events from bedside monitors of three manufacturers in US hospitals, with at least two expert annotators per event and

the alarm at the 300 s mark of each recording [4], [13]. The database licence is CC BY-SA 4.0. Version 1.1 also releases the individual annotator votes. The host throttles every connection, so only the window from 60 s before to 30 s after each alarm was downloaded with parallel range requests, about 1 GB instead of 18 GB. After removing 292 events with no pulse channel and 203 low-quality windows, 4,542 events from 2,138 patients remained, 28.0% of them true alarms. The official patient-disjoint split gives 3,662 training, 453 validation and 427 test events, with 120 true alarms in the test split. Hospital and device identifiers are not released.

3.3 Data quality, imbalance and splits

Quality control was crude. Each window was judged on its own samples, and windows that were too short, almost flat or clipped, or that scored below 0.05 on our quality measure, were dropped. The counts above say how many. Each Challenge 2015 record yields one window, so splitting by record and by window comes to the same thing. We did not look for duplicates across records beyond the record identifiers. Class imbalance (about 38% true alarms in Challenge 2015 and 28% in VTaC) is handled in the loss function and not by resampling, because resampling can put copies of one patient on both sides of a split.

Challenge 2015 was evaluated with record-wise 5-fold cross-validation repeated for 10 seeds, which gives folds of about 119 test records. In each fold, 15% of the remaining records form an inner validation set (about 71 records), leaving about 402 for fitting. VTaC uses the official split. The test fold or test split was used once per model to produce the numbers in Section 7. No augmentation or resampling was applied to validation or test data.

3.4 Auxiliary data

MIMIC PERform AF holds 20-minute recordings of 35 critically ill adults at 125 Hz, 19 with atrial fibrillation and 16 without [10], [14]. Sixty evenly spaced 10 s windows per subject (2,100 in total) were used for a side task. Real noise for the stress tests came from the MIT-BIH Noise Stress Test database [15]. Each dataset’s licence allows research use, and redistribution of derived windows should follow the stricter of the PhysioNet terms and CC BY-SA 4.0. Patient overlap between the corpora cannot be checked from the released identifiers.

4. Proposed Methodology

4.1 Overall approach

We did not start with a finished design. We started with one hypothesis, that cross-attention between ECG and pulse would verify alarms better than simpler fusion, and then spent most of the project trying to break it. The cross-attention network came first, with three relatives of the same size (ECG-only, PPG-only, concatenation) so that any difference could be traced to the fusion step. Next came the rivals: other fusion layers, other network families, feature-based learners and a stronger training recipe. The last stage was to perturb the inputs and see what the networks actually use.

4.2 Workflow

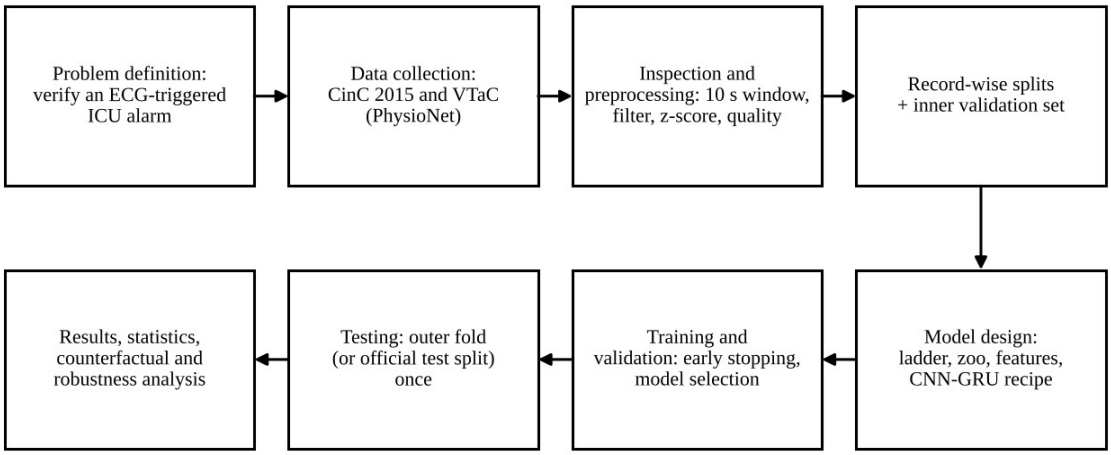


Figure 1: Workflow of the project. Records are split before any model is fitted, and the outer test data are used once.

4.3 Preprocessing

Signals were resampled to 250 Hz, then band-pass filtered with a zero-phase fourth-order Butterworth filter (ECG 0.5 to 40 Hz, pulse 0.5 to 8 Hz). Each 10 s window was z-normalised on its own samples, which removes absolute amplitude differences caused by gain and placement. The window is aligned to end at the alarm onset. An earlier version centred the window on the alarm, which let the model see samples after it; that was corrected, and the 30 s after the alarm are now used only in a separate time-to-verdict analysis. For the classical models, 266 features were computed per window and imputed, clipped and standardised using statistics from the fitting data of each fold only.

4.4 Model selection

We built cross-attention first because the project hypothesis was about alignment between two signals. The ladder variants were scaled to about 261k parameters each, by adjusting channel widths, so a gap between them cannot be blamed on capacity. The wider comparison covers four other fusion layers (bidirectional cross-attention, joint transformer, bottleneck tokens, quality-gated cross-attention) and six early-fusion families: ResNet1D [17], InceptionTime [18], a bidirectional GRU, a temporal convolutional network (TCN) [19], a patch transformer and a small state-space model. Together with the four ladder variants that is 14 neural architectures. Classical learners (logistic regression, RBF-SVM, random forest, extra trees, XGBoost [20] and an MLP) ran on engineered features, with a Pan-Tompkins-style beat detector [21] behind the timing features. We expected the feature route to be strong with only a few hundred records.

The strongest recipe came from a grid declared before any run: six architectures, augmentation on or off, two losses and two learning rates, 48 configurations in all, scored only on inner-validation AUC. An attention CNN-GRU, following the design of Mousavi et al. as far as the published description allows, won the grid

and was then combined with weight averaging, test-time augmentation and a three-member ensemble.

4.5 Implementation details

Everything is Python 3. The models use PyTorch [22]; the classical models and metrics use scikit-learn [23] and XGBoost; signal processing uses SciPy, PyWavelets and pandas. We ran the experiments on one workstation with an RTX PRO 4000 Blackwell GPU (24 GB) and 20 CPU cores. The GPU was shared, which is why the recipe uses epoch caps and short patience. A cross-attention fold trains in about 5 s on average, a recipe fold (three members) in about 21 s on Challenge 2015 and 45 s on VTaC. Annexure A lists the repository, which holds the code, configuration files and every result file used here.

5. Model Architecture

5.1 Cross-attention fusion network (the proposed model)

The proposed model gives each signal its own one-dimensional CNN. The weights are not shared, because ECG and pulse look and fail so differently. The encoders do not pool the time axis away, since attention needs a sequence to compare. ECG positions act as queries over the pulse positions, which supply the keys and values, and the fused sequence is averaged over time before a small classification head.

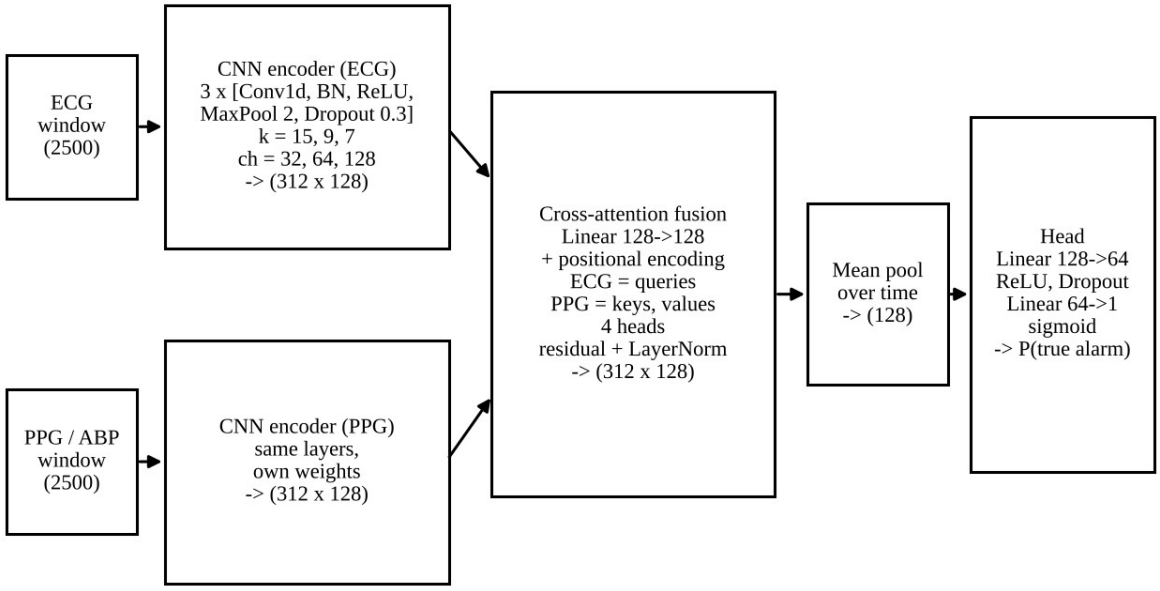


Figure 2: Cross-attention fusion network. Shapes are (time steps x channels), and the two encoders have separate weights.

Table 4: Layer-by-layer description of the cross-attention fusion network (width multiplier 1.0; 261,505 trainable parameters).

Layer	Output shape	Role
Input (ECG and PPG/ABP)	2 x (2500)	Two 10 s windows at 250 Hz, z-normalised

Layer	Output shape	Role
Conv1d(1 to 32, k = 15), BN, ReLU, MaxPool 2, Dropout 0.3	(1250 x 32)	Local waveform features, such as QRS shape or pulse upstroke
Conv1d(32 to 64, k = 9), BN, ReLU, MaxPool 2, Dropout 0.3	(625 x 64)	Longer-range patterns
Conv1d(64 to 128, k = 7), BN, ReLU, MaxPool 2, Dropout 0.3	(312 x 128)	Final sequence of 312 feature vectors per signal
Linear(128 to 128) + sinusoidal positional encoding	(312 x 128)	Gives attention a sense of time order
Multi-head attention, 4 heads, dimension 128; ECG queries, PPG keys and values	(312 x 128)	Each ECG step collects the pulse steps it finds relevant
Residual connection + LayerNorm	(312 x 128)	Stabilises the fused sequence
Mean over time	(128)	Summary vector for the whole window
Linear(128 to 64), ReLU, Dropout 0.3, Linear(64 to 1), sigmoid	(1)	Probability that the alarm is true

The ECG-only and PPG-only models use one encoder and a mean-pooled head, with channel widths scaled by 1.79 to reach the same parameter count. The concatenation model keeps both encoders but replaces attention with a linear projection of each pooled signal followed by concatenation. Details are in the configuration files of the repository. The sigmoid output is a probability, and the loss and optimiser are described in Section 6.

5.2 Attention CNN-GRU recipe (the strongest deep model)

This model replaces cross-attention with a recurrent branch per signal. A three-block CNN shrinks 2,500 samples to 39 time steps, a bidirectional GRU reads them, and additive attention pools the sequence into one context vector. The two context vectors are concatenated and classified. Three members with different initialisations are trained and their probabilities averaged.

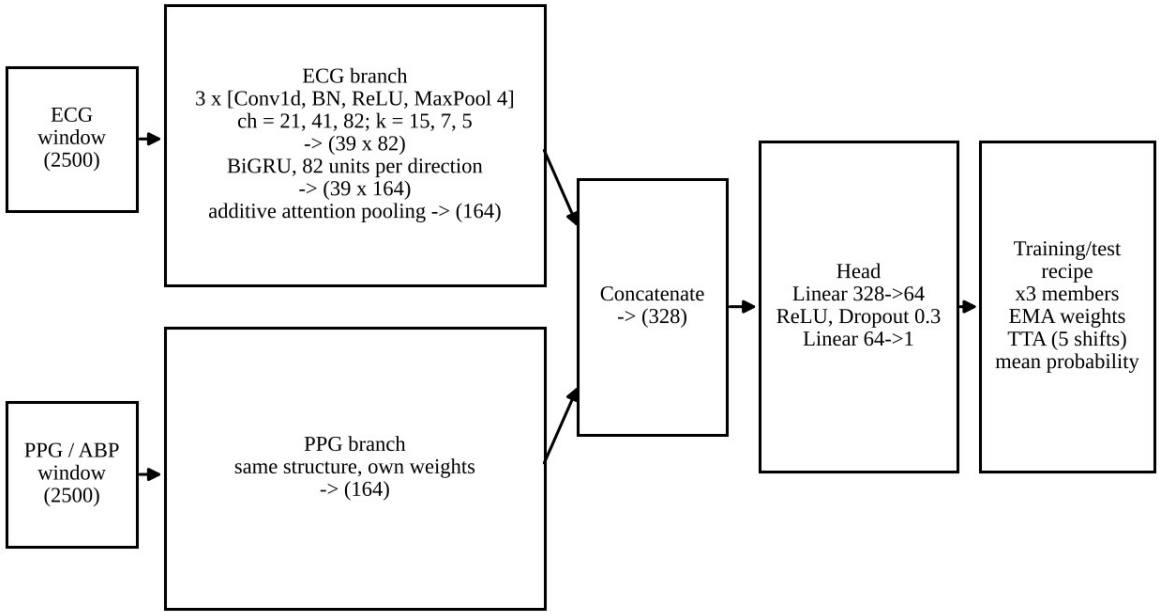


Figure 3: Architecture of the attention CNN-GRU recipe. Per-signal branches do not share weights; three members are averaged at test time.

Table 5: Layer-by-layer description of one member of the recipe (width multiplier 1.287; 258,857 trainable parameters).

Layer (per signal branch)	Output shape	Role
Conv1d(1 to 21, k = 15), BN, ReLU, MaxPool 4	(625 x 21)	Waveform features, coarse time step
Conv1d(21 to 41, k = 7), BN, ReLU, MaxPool 4	(156 x 41)	Intermediate features
Conv1d(41 to 82, k = 5), BN, ReLU, MaxPool 4	(39 x 82)	Sequence of 39 steps
Bidirectional GRU, 82 units per direction, Dropout 0.3	(39 x 164)	Temporal context in both directions
Additive attention pooling (dimension 82)	(164)	Weighted summary of the window
Concatenate ECG and PPG branches	(328)	Joint representation
Linear(328 to 64), ReLU, Dropout 0.3, Linear(64 to 1)	(1)	Logit; sigmoid gives the probability

Each branch has 118,868 parameters and the head has 21,121, which adds up to the reported 258,857. Training uses augmentation (a joint time shift of up to one second, amplitude scaling, noise and random dropout of one modality), focal loss, AdamW with a one-cycle schedule, an exponential moving average of the weights and, at test time, averaging over five small time shifts.

5.3 Transfer learning with a PPG foundation model

A side experiment tested whether an open PPG foundation model could replace the scratch PPG encoder. The pretrained network was PaPaGei-S (a ResNet1D

with a mixture-of-experts, 512-dimensional embeddings) [11], with weights from Zenodo record 13983110 and the authors' code (BSD-3-Clause-Clear licence; the weights record states no licence). Its input is 10 s at 125 Hz, so windows were decimated from 250 Hz. Three setups were run: a frozen encoder with a linear probe, a frozen encoder feeding a scratch ECG CNN and the same fusion layers, and fine-tuning of only the last residual block and final batch normalisation at a learning rate of 1e-4 while all other layers stayed frozen. Low-rank adapters were not used.

6. Mathematical Model

Only the mathematics that the code uses is given here.

6.1 Activation and output

Every convolution is followed by batch normalisation and the rectified linear unit

$$f(x) = \max(0,\ x) \tag{1}$$

The last layer outputs a logit z, and the sigmoid below converts it to the probability that the alarm is true. Section 7 uses a threshold of 0.5 for the class labels.

$$p = \sigma(z) = \frac{1}{1 + e^{-z}} \tag{2}$$

6.2 Cross-attention

E and P are the ECG and PPG feature sequences after the linear projection and positional encoding, each of shape (T' x d), and W_Q, W_K and W_V are learned matrices. The attention output is

$$\mathrm{Attn}(E, P) = \mathrm{softmax}\left(\frac{(EW_Q)(PW_K)^{\top}}{\sqrt{d_k}} \right) PW_V \tag{3}$$

Here d_k is the key dimension, 128 divided by the 4 heads. Row i of the softmax holds the weights with which ECG step i reads the pulse steps. A network that tracked pulse transit time would put these weights at a lag matching the physiological delay. Ours did not (Section 8).

6.3 Recurrent layer and attention pooling

The recurrent layer in the recipe is a gated recurrent unit, with update gate z_t, reset gate r_t and candidate state h~_t:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z), \quad r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$

$$\tilde{h}_t = \tanh(W x_t + U(r_t \odot h_{t-1}) + b), \quad h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \tag{4}$$

Sigma is the sigmoid, the circled dot is element-wise multiplication, and W, U and b are learned. The GRU runs forwards and backwards over the 39 steps,

giving states h_t of size 164, and additive attention pools them into one context vector c:

$$e_t = v^{\top} \tanh(W h_t + b), \quad a_t = \frac{\exp(e_t)}{\sum_s \exp(e_s)}, \quad c = \sum_t a_t h_t \tag{5}$$

6.4 Loss function

Both model families train with a focal loss [24]. It is a weighted binary cross-entropy in which each example is multiplied by (1 minus p_t) raised to a power gamma, with gamma = 2. Here p_t is the probability the model gives to the correct class, and w is a positive-class weight for the imbalance:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (1 - p_{t,i})^{\gamma} \ell_i, \quad \ell_i = -[w\, y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad \gamma = 2 \tag{6}$$

Examples the model already gets right (p_t near 1) count for little, so training spends its effort on the hard ones. Most alarms are easy and the rare true alarms matter most, which is why we chose it. The recipe grid also tried binary cross-entropy with label smoothing of 0.05.

6.5 Optimiser and weight averaging

The baseline ladder trains with Adam. The recipe uses AdamW [25], which applies weight decay lambda directly to the weights and not through the gradient, together with a one-cycle learning-rate schedule [26] (15% warm-up, then cosine decay). With gradient g_t, moment decays beta_1 and beta_2 and bias-corrected moments, one step is

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad \theta_t = \theta_{t-1} - \eta_t \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} + \lambda\, \theta_{t-1} \right) \tag{7}$$

The recipe also keeps an exponential moving average of the weights (horizon of about three epochs), and the averaged weights at their best inner-validation epoch are used for prediction:

$$\bar{\theta}_t = \beta\, \bar{\theta}_{t-1} + (1 - \beta)\, \theta_t \tag{8}$$

6.6 Regularisation

Dropout (rate 0.3) is applied after each convolution block and in the head; weight decay is 1e-4 for the ladder and 1e-2 for the recipe; early stopping on inner-validation data ends training when the validation score stops improving. Time-shift, scaling, noise and modality-dropout augmentation act as further regularisers in the recipe.

7. Results and Experiments

7.1 Experimental setup

All experiments ran on the workstation described in Section 4.5. For Challenge 2015 each seed gives five folds with about 402 fitting, 71 inner-validation and 119 test records; seeds 0 to 9 vary the fold assignment and the initialisation (50 test evaluations per model). For VTaC the official split gives 3,662 training, 453 validation and 427 test events (120 true alarms), and seeds 0 to 4 vary the initialisation (10 seeds for the two single-signal models). The recipe grid used seeds 100 to 102 on the Challenge 2015 PPG cohort and never produced test predictions. The configuration file sets a global seed of 42. All models run on the same splits, which makes the comparisons paired.

7.2 Hyperparameter details

Most settings were fixed in advance and not tuned. The table states which ones were searched and which were simply set.

Table 6: Hyperparameters: values searched and final values.

Parameter	Values tested	Final value
Learning rate	Recipe grid: 3e-4 and 1e-3. Ladder: not tuned	1e-3 (ladder and recipe)
Batch size	Not tuned	16 (ladder); 32 (recipe, Challenge 2015); 64 (recipe, VTaC)
Epochs	Not tuned; early stopping	Ladder: up to 60, patience 10. Recipe: up to 30, patience 8 (Challenge 2015); up to 20, patience 6 (VTaC)
Optimiser	Not compared	Adam (ladder); AdamW with one-cycle schedule (recipe)
Dropout and weight decay	Not tuned	Dropout 0.3; weight decay 1e-4 (ladder), 1e-2 (recipe)
Loss	Recipe grid: focal loss and BCE with label smoothing 0.05	Focal loss, gamma = 2
Augmentation	Recipe grid: off and on	On
Architecture	Recipe grid: 6 architectures	Attention CNN-GRU (mean inner-validation AUC 0.936 for the winning configuration)
Ensemble size	Ablation: 1, 2 and 3 members	3
Classical models	30 random configurations per fold and algorithm, chosen on inner validation	Chosen separately in each fold

Every ladder variant used the same settings; none was tuned separately. The recipe grid was scored only on inner-validation AUC, and the test folds were not used to make any of these choices, with the one caveat discussed in Section 8.

7.3 Performance metrics

The positive class is "true alarm". Accuracy, precision, recall (sensitivity) and F1 follow the standard definitions, where TP, TN, FP and FN are true positives, true

negatives, false positives and false negatives. Specificity is TN/(TN+FP), and AUC is the area under the ROC curve.

$$\mathrm{Acc} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \mathrm{Prec} = \frac{TP}{TP + FP}, \quad \mathrm{Rec} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2\,\mathrm{Prec}\,\mathrm{Rec}}{\mathrm{Prec} + \mathrm{Rec}} \tag{9}$$

We also report the PhysioNet Challenge score [7], in which one missed true alarm costs as much as five false alarms:

$$S = \frac{TP + TN}{TP + TN + FP + 5\,FN} \tag{10}$$

To compare models we used a paired record-level bootstrap with 5,000 resamples [27], the Nadeau-Bengio corrected resampled t-test [28] and Holm correction [29]. The comparison family was written into the configuration file before the final runs. A difference counts as robust only when both tests agree.

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{J} + \frac{n_{\mathrm{test}}}{n_{\mathrm{train}}}\right) s_d^2}} \tag{11}$$

7.4 Tables and graphs

Main comparison

Tables 7 and 8 compare the models on the held-out folds and the held-out test split. Values are means over folds and seeds, with the standard deviation across them for F1 and AUC; these standard deviations are not confidence intervals, since folds share training data.

Table 7: Challenge 2015, PPG cohort (592 records; 5-fold cross-validation, 10 seeds, 50 test folds).

Model	Acc.	F1	AUC	Sens.	Spec.
ECG-only CNN	0.708	0.642 ± 0.063	0.782 ± 0.049	0.704	0.714
PPG-only CNN	0.611	0.602 ± 0.059	0.719 ± 0.051	0.799	0.499
CNN concatenation	0.757	0.693 ± 0.052	0.836 ± 0.040	0.731	0.773
CNN cross-attention (proposed)	0.765	0.699 ± 0.049	0.841 ± 0.033	0.734	0.786
ResNet1D	0.820	0.760 ± 0.060	0.884 ± 0.039	0.764	0.854
TCN	0.810	0.751 ± 0.053	0.887 ± 0.034	0.765	0.838
InceptionTime	0.796	0.732 ± 0.068	0.876 ± 0.044	0.745	0.827
Gradient boosting, features v1	0.851	0.798 ± 0.040	0.918 ± 0.025	0.788	0.891
Attention CNN-GRU recipe	**0.871**	**0.828 ± 0.037**	**0.932 ± 0.028**	**0.827**	**0.898**
XGBoost, features v2	0.871	0.818 ± 0.049	0.939 ± 0.023	0.782	0.925
Extra Trees, features v2	0.867	0.802 ± 0.047	0.947 ± 0.020	0.725	0.953
Logistic regression, features v2	0.852	0.801 ± 0.043	0.928 ± 0.021	0.799	0.886
MLP, features v2	0.857	0.802 ± 0.047	0.927 ± 0.023	0.778	0.906

Table 8: VTaC, official split (427 test events, 120 true alarms; 5 seeds, 10 for the single-signal models).

Model	Acc.	F1	AUC	Sens.	Spec.
ECG-only CNN	0.824	0.709 ± 0.029	0.897 ± 0.008	0.769	0.845
PPG-only CNN	0.693	0.566 ± 0.027	0.751 ± 0.014	0.712	0.685
CNN concatenation	0.837	0.744 ± 0.013	0.909 ± 0.008	0.838	0.837
CNN cross-attention (proposed)	0.844	0.746 ± 0.013	0.915 ± 0.012	0.813	0.857
CNN bidirectional cross-attention	0.857	0.775 ± 0.019	0.933 ± 0.008	0.878	0.848
ResNet1D	0.852	0.750 ± 0.023	0.915 ± 0.010	0.792	0.876
TCN	0.856	0.760 ± 0.020	0.919 ± 0.011	0.813	0.873
Attention CNN-GRU recipe	**0.913**	**0.858 ± 0.008**	**0.966 ± 0.001**	**0.933**	**0.905**

Ensembles of ten models reached accuracy 0.876, F1 0.827 and AUC 0.948 on Challenge 2015 (10 seeds), and 0.910, 0.839 and 0.955 on VTaC (5 seeds). Adding the recipe to the ensemble gave accuracy 0.878, F1 0.830 and AUC 0.950 on Challenge 2015 (5 seeds). The full leaderboard with bootstrap intervals is in the repository file docs/LEADERBOARD.md.

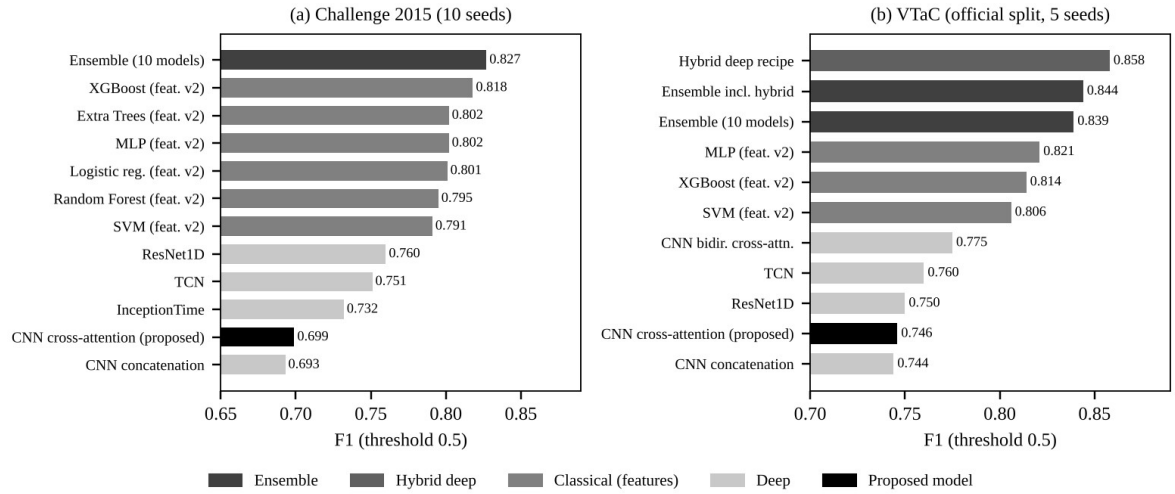


Figure 4: F1 of the compared models on Challenge 2015 (left) and VTaC (right), threshold 0.5. The originally proposed cross-attention model (black) is near the bottom on both.

Confusion matrices and class-wise reports

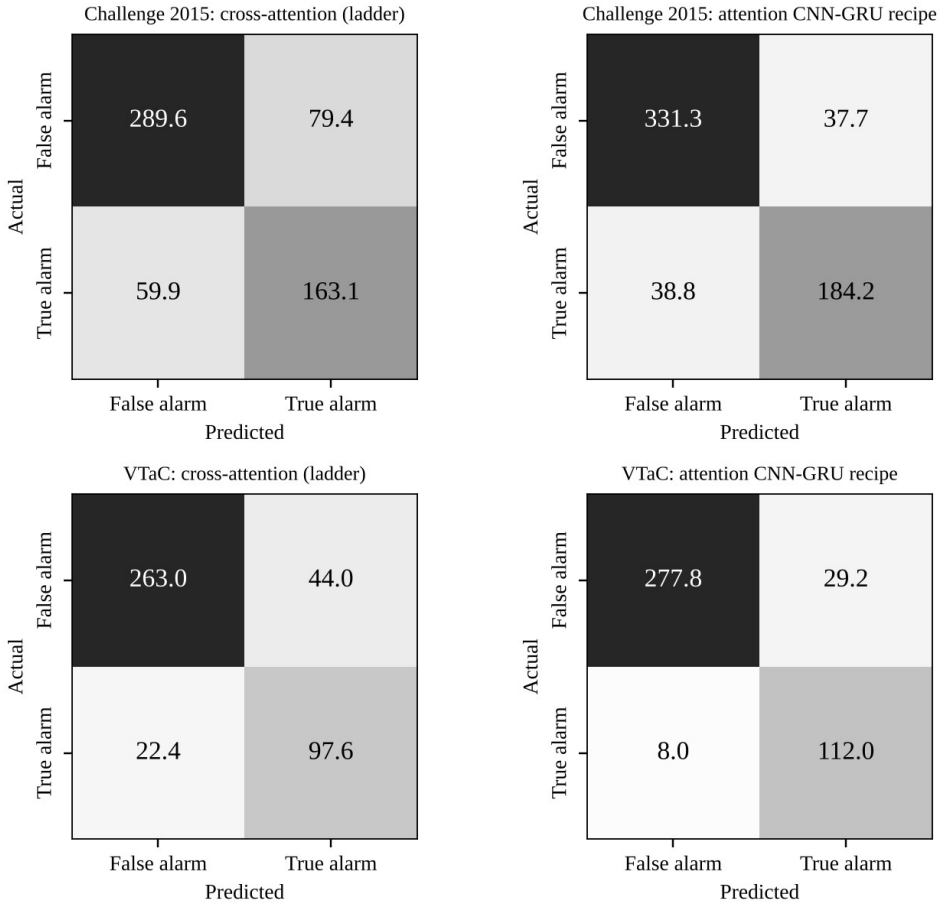


Figure 5: Confusion matrices, as counts per test pass (Challenge 2015: pooled over 50 test folds and divided by 10 seeds, 592 records; VTaC: mean of 5 seeds, 427 events).

Confusion matrices were built from the true-positive, true-negative, false-positive and false-negative counts stored for every fold. The class-wise numbers below come from the pooled counts, so they differ in the third decimal from the fold-averaged values in the previous tables.

Table 9: Class-wise classification report from the pooled confusion counts (support is per seed).

Model and dataset	Class	Precision	Recall	F1	Support
Cross-attention, Challenge 2015	True alarm	0.673	0.731	0.701	223
	False alarm	0.829	0.785	0.806	369
	Macro avg. / weighted avg. F1	0.751 / 0.770	0.758 / 0.765	0.753 / 0.766	592
CNN-GRU recipe, Challenge 2015	True alarm	0.830	0.826	0.828	223
	False alarm	0.895	0.898	0.896	369
	Macro avg. / weighted avg. F1	0.863 / 0.871	0.862 / 0.871	0.862 / 0.871	592

Model and dataset	Class	Precision	Recall	F1	Support
Cross-attention, VTaC	True alarm	0.689	0.813	0.746	120
	False alarm	0.922	0.857	0.888	307
	Macro avg. / weighted avg. F1	0.805 / 0.856	0.835 / 0.844	0.817 / 0.848	427
CNN-GRU recipe, VTaC	True alarm	0.793	0.933	0.858	120
	False alarm	0.972	0.905	0.937	307
	Macro avg. / weighted avg. F1	0.883 / 0.922	0.919 / 0.913	0.897 / 0.915	427

Training behaviour

Per-epoch training and validation curves were not archived by the training scripts: each run stored the epoch of the best validation score, the validation AUC at that epoch and the final test metrics, and the model checkpoints were not kept. Plots of accuracy and loss against epoch can therefore not be shown without re-running the training, and none has been drawn here. What the stored records do show is where early stopping ended training.

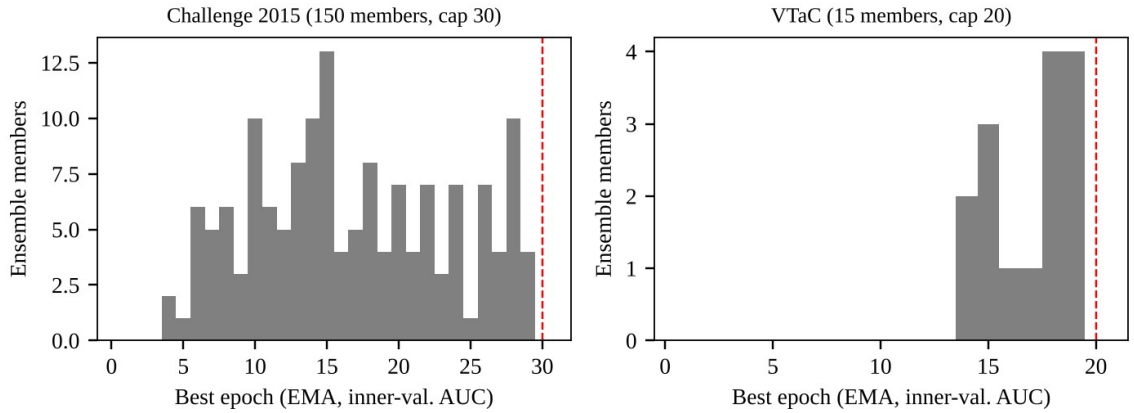


Figure 6: Epoch at which each ensemble member of the CNN-GRU recipe reached its best inner-validation AUC (EMA weights). Dashed line: epoch cap.

On Challenge 2015 the best epoch is spread widely between 4 and 29 with a median region around 15, so most members stopped well before the 30-epoch cap. On VTaC all 15 members peaked between epoch 14 and 19 of a 20-epoch cap, which suggests those runs were still improving slowly when the cap arrived and might gain a little from more epochs.

Effect of the validation leak

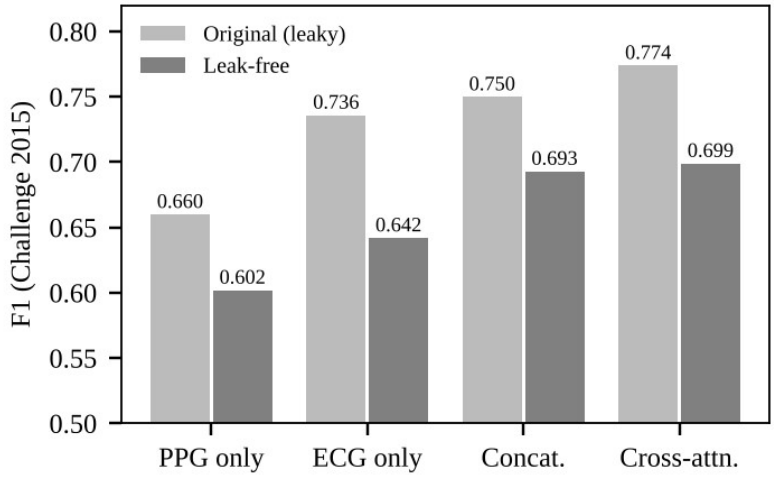


Figure 7: F1 on Challenge 2015 before and after the validation leak was removed. "Original" scores selected the best epoch on the fold that was then reported.

The first version of the training script chose the best epoch and the stopping point using the same fold it then reported. Fixing that, with a separate inner validation set and 10 instead of 3 seeds, lowered every F1: PPG-only 0.660 to 0.602, ECG-only 0.736 to 0.642, concatenation 0.750 to 0.693 and cross-attention 0.774 to 0.699. The cross-attention lead over concatenation shrank from +0.024 to +0.006. The original numbers are archived in the repository and are not used in any other table.

Representative predictions

Two persistent errors of the cross-attention model are shown below, with the ECG (top) and PPG (bottom) windows, the detected beats and the model outputs. The text above each figure is produced by the error-analysis script; the note under the plot is an automatic heuristic, not a clinician annotation.

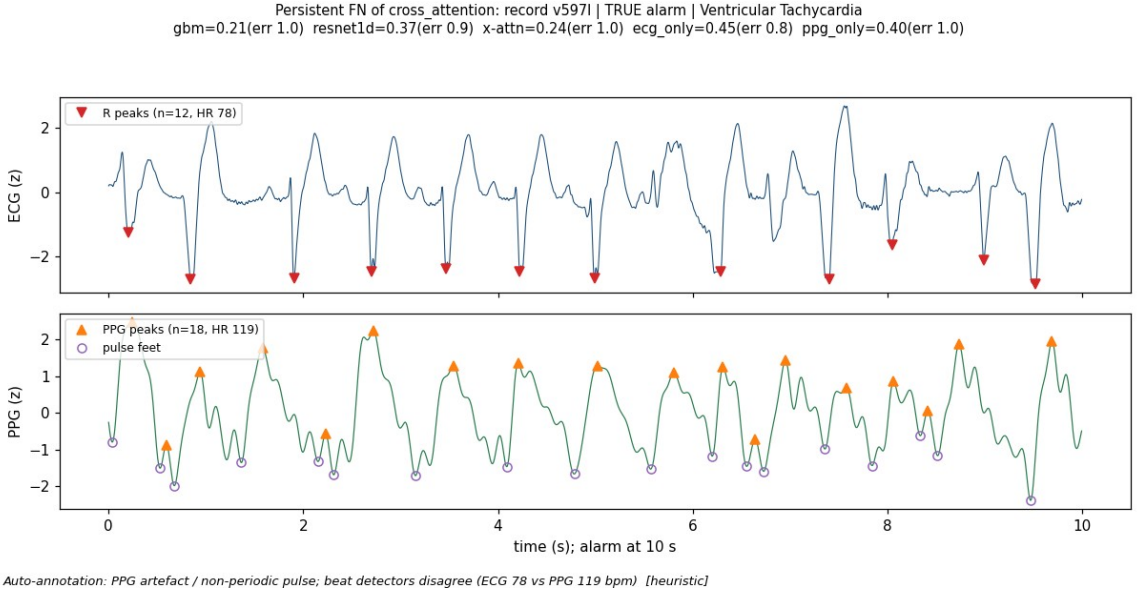


Figure 8: A missed true alarm (false negative). Record v597l is a true ventricular tachycardia alarm. The cross-attention model gives it a true-alarm probability of 0.24, and the gradient-

boosting and ResNet1D models also miss it (0.21 and 0.37). The automatic note under the plot calls the PPG non-periodic and says the two beat detectors disagree (ECG 78 bpm, PPG 119 bpm).

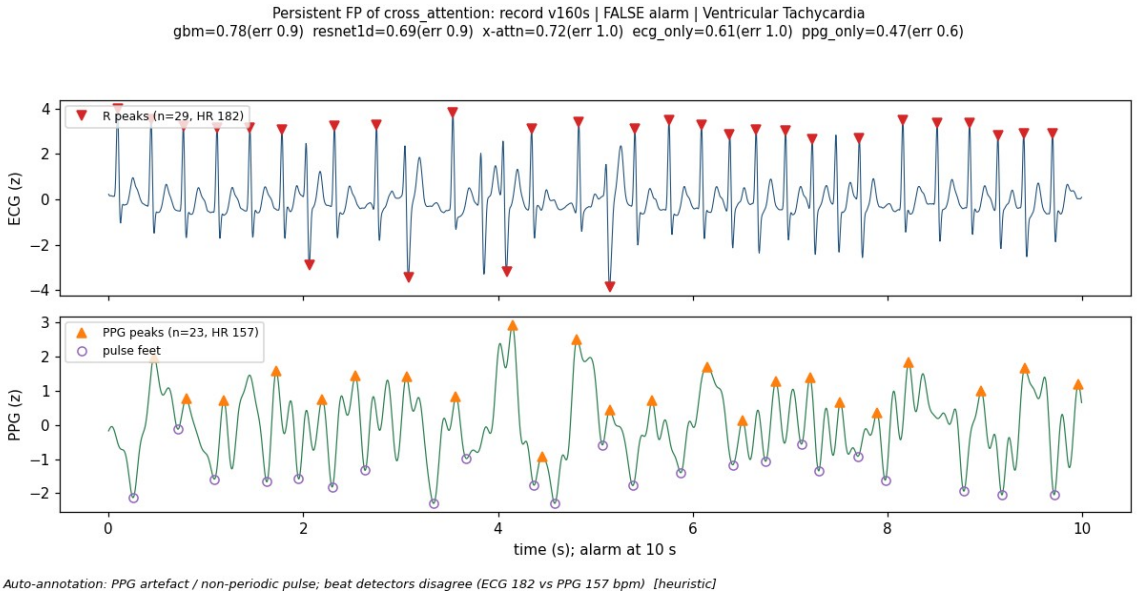


Figure 9: A false alarm accepted as true (false positive). Record v160s is a false ventricular tachycardia alarm that cross-attention scores 0.72, and the gradient-boosting and ResNet1D models also score it above 0.5 (0.78 and 0.69). The ECG shows a fast, regular run of 29 detected beats (182 bpm) and the PPG has an irregular shape (157 bpm).

Both examples make the same point from opposite sides. In the first, the ECG alone looks like a calm rhythm and the noisy pulse is misleading; in the second, a plausible-looking fast ECG run is enough to convince the model. Neither case contains a clean disagreement between the two signals of the kind the original design hoped to exploit.

8. Results Discussion

The first thing the numbers say is that the proposed model did not win. On Challenge 2015 the cross-attention network scores F1 0.699, below every feature-based model and below ResNet1D, TCN and InceptionTime in Table 7. Only concatenation and the two single-signal models score lower. Against concatenation, which has the same encoders and no attention, it gains 0.006 F1, with a 95% bootstrap interval from -0.015 to +0.028 [27]. Four attention variants (bidirectional, joint transformer, gated, and the plain one) cannot be told apart. Fusion itself does help: on VTaC, ECG-only reaches F1 0.709 and PPG-only 0.566 against 0.744 for concatenation. It is the attention part that adds nothing measurable.

The strongest results came from other routes. XGBoost on engineered features reached F1 0.818 and AUC 0.939 on Challenge 2015, and the attention CNN-GRU recipe reached F1 0.828 there and 0.858 (AUC 0.966) on VTaC. The ten-model ensemble's F1 lead over the best classical model on Challenge 2015 (+0.008, interval -0.007 to +0.022) is inside the noise, while its lead over cross-attention (+0.128) is not. On VTaC the top three models are within the bootstrap

interval of each other, but all three are well above cross-attention (+0.11 F1 for the recipe).

The confusion matrices show where the errors sit. On VTaC the recipe misses 8 of 120 true alarms per seed on average (recall 0.933) and raises about 29 false positives among 307 false alarms (specificity 0.905); cross-attention misses about 22 true alarms and raises 44 false ones. On Challenge 2015 the recipe misses about 39 of 223 true alarms, against 60 for cross-attention. Part of this comes from alarm type. Cross-attention scores F1 0.92 on tachycardia, but its specificity there is only 0.42: 106 of the 111 tachycardia alarms are true, and the model rarely says no. VT gets an F1 of 0.49. Asystole (14 true alarms) and flutter/fibrillation (5) are too small to judge. In the error analysis VT also dominates the persistent misses, 18 of 29 for the gradient-boosting model, although VT makes up only 26% of true alarms.

Without training curves, overfitting can only be judged indirectly. For the recipe, the chosen members had a mean inner-validation AUC of 0.931 on Challenge 2015 and 0.965 on VTaC, and the test AUCs were 0.932 and 0.966, so we see no large validation-to-test gap. That check is soft. Validation numbers are maxima over epochs and so slightly optimistic, and we are comparing single members with an ensemble. For the ladder models the picture is murkier: test scores are modest and validation sets are small, so we cannot separate underfitting from a weak use of the second signal. A Challenge 2015 validation set holds only 15 to 25 true alarms, which makes epoch selection noisy.

The comparison is fair on splits, seeds and parameter count, and unfair on training. The ladder variants use Adam, F1-based early stopping and a batch of 16. The recipe adds augmentation, AdamW, weight averaging and ensembling. The ablation shows what that is worth: dropping weight averaging costs 0.008 F1, a single member with raw weights and no test-time augmentation loses 0.035, and going from one member to three adds about 0.012. So part of the gap between the recipe and the baseline CNNs belongs to the trainer, not the architecture. We looked for leakage in the recipe and found none: a shuffled-label control on one fold gave an AUC of 0.57, which is chance, and a split check confirmed that fitting, validation and test records were disjoint. One bias remains. We chose the recipe configuration on inner-validation scores from Challenge 2015 folds, and those records are test records in other seeds, so the numbers for that dataset may be slightly optimistic. The VTaC result, where the same recipe ran without re-tuning, is the cleaner check.

Why does the fusion network not use timing? Replacing the pulse with another patient’s pulse of the same label changed AUC by only 0.002 for cross-attention and 0.015 for concatenation, even though that pulse has no timing relation to the ECG. Shifting the pulse by up to two seconds changed AUC by at most 0.005 for the CNN fusion models. An opposite-label pulse, in contrast, dropped cross-attention AUC from 0.842 to 0.535. The network reads the pulse as a patient-independent cue (is there a regular, well-formed pulse or not), not as an answer to "did this beat produce a pulse". The attention weights agree: the lag they imply has a Spearman correlation of -0.039 with the measured pulse transit time (permutation $p = 0.39$), and only 5.3% of the attention mass falls in the physiological lag range, against 5.6% for uniform attention. The pulse-transit-time argument that motivated the design is not supported by these data.

Occlusion-based explanations were faithful in deletion and insertion tests while attention weights were barely better than a random ordering, so attention maps should not be presented as explanations.

The noise tests point the same way. We added real Noise Stress Test noise to the ECG and synthetic motion artefact to the PPG. At 0 dB signal-to-noise ratio on both, ResNet1D held its AUC at 0.844 (0.907 clean), while cross-attention fell from 0.856 to 0.696 and concatenation from 0.833 to 0.720. Zeroing the PPG at test time cost cross-attention 0.12 AUC and ResNet1D 0.27. Transfer between datasets is lopsided. Models trained on Challenge 2015 scored an F1 of only 0.56 to 0.60 on VTaC when they used the ECG (PPG-only was near chance at 0.432). Models trained on VTaC did well on the VT alarms of Challenge 2015: ResNet1D reached F1 0.773 and AUC 0.927. More data helped more than a cleverer model, and numbers from one dataset overstate what a model would do elsewhere.

Strengths of the work are the controlled protocol, a leak that was found and corrected, and perturbation tests that probe what the networks use. Its limits are real. The datasets are small and the fold-to-fold F1 standard deviation is about 0.05. The engineered features and, on Challenge 2015, the alarm type itself carry the alarm definitions, which is built-in knowledge and not learned physiology; the alarm-type ablation was not run. The reproduction of the published attention CNN-RNN is ours and had only a two-seed sanity run outside the recipe. There was no prospective validation, no demographic analysis of optical PPG, and a safety layer (calibration and a certified suppression threshold) was run only for the concatenation model, where validation-chosen thresholds aimed at 95 to 100% sensitivity realised about 92 to 97% on test folds, so sensitivity targets should not be assumed to transfer.

9. Conclusion and Future Scope

The project set out to show that cross-attention between ECG and pulse verifies ICU arrhythmia alarms better than simpler models. After a validation leak was removed it did not. The cross-attention network reached F1 0.699 and AUC 0.841 on Challenge 2015 and F1 0.746 on VTaC, matching naive concatenation and trailing feature-based models, ResNet1D and the attention CNN-GRU recipe, which reached F1 0.828 on Challenge 2015 and F1 0.858 with accuracy 0.913 and AUC 0.966 on the VTaC test split. The objectives were met in the sense that all five were carried out and reported, but the central hypothesis was not confirmed.

What the project contributes is the evidence behind that result: matched-size comparisons, paired statistics, counterfactual tests showing that the networks use signal content and not ECG-to-pulse timing, and an explicit record of the leak that had inflated our first numbers by 0.075 F1. The weak points are the small datasets, the lack of prospective validation, the poor transfer from Challenge 2015 to VTaC, and the training curves we did not save.

Realistic next steps are to pretrain on larger unlabelled pulse and ECG corpora (PulseDB, VitalDB and MIMIC-III were not accessible), to test leave-one-site-out and leave-one-device-out, to try a window that keeps the pre-alarm 10 s and adds post-alarm seconds, to rerun key models while logging per-epoch curves, to run the alarm-type ablation, and eventually to evaluate prospectively with

clinician review. Any deployment would need the model to assist, not replace, clinical judgement.

10. References

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Annexure A: Code Repository

Repository: https://github.com/VeerbhadraMahant/ecg-ppg-pipeline

The README describes the pipeline and the project layout. Python dependencies are listed in requirements.txt (numpy, scipy, pandas, scikit-learn, matplotlib, torch, wfdb, pyyaml, tqdm, xgboost, lightgbm, PyWavelets and the Jupyter tools, among others). The steps below reproduce the main results; the datasets are downloaded separately from PhysioNet and Zenodo and are not stored in the repository.

- Create an environment and install the packages: python -m venv .venv, activate it, then pip install -r requirements.txt.
- Download the data: python scripts/download_data.py --dataset challenge2015 and python scripts/download_vtac.py --workers 12.
- Preprocess: python -m src.data.preprocess, then python scripts/build_vtac_windows.py.
- Match parameter counts: python scripts/match_params.py.
- Train the baseline ladder with 10 seeds: python -m src.train --variant all --run-name challenge2015_ppg; for VTaC use --protocol official with the VTaC windows file.
- Classical models: python -m src.classical --dataset cinc --run-name classical_cinc.
- Evaluate and tabulate: python -m src.evaluate --run-name challenge2015_ppg and python scripts/final_leaderboard.py.
- Unit tests: python -m pytest tests.
- Regenerate the figures of this report from the stored results: python report/make_report_figures.py; rebuild this document with node report/build_report.js.

Annexure B: Additional Results

B.1 Paired comparisons

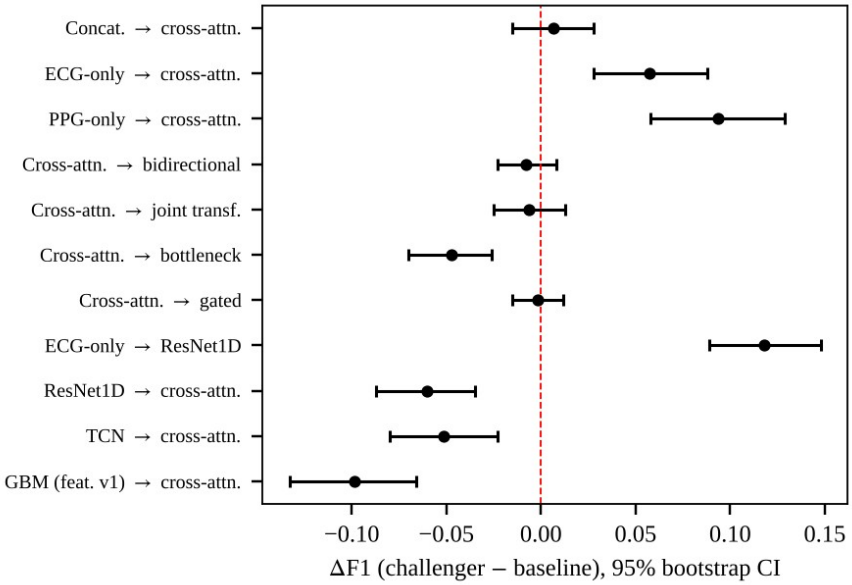


Figure 10: Paired F1 differences (challenger minus baseline) with 95% bootstrap intervals on Challenge 2015, from runs/challenge2015_ppg/significance.csv.

Table 10: Paired F1 differences on Challenge 2015 (50 seed-by-fold pairs); Holm-adjusted p-values.

Baseline	Challenger	dF1	95% CI	p (N-B)	p (boot.)
Concatenation	Cross-attention	+0.006	[-0.015, +0.028]	1.00	1.00
ECG-only	Cross-attention	+0.058	[+0.028, +0.088]	1.00	0.003
PPG-only	Cross-attention	+0.094	[+0.058, +0.129]	0.077	< 0.001
Cross-attention	Bidirectional	-0.008	[-0.023, +0.008]	1.00	1.00
Cross-attention	Joint transformer	-0.006	[-0.025, +0.013]	1.00	1.00
Cross-attention	Gated	-0.002	[-0.015, +0.012]	1.00	1.00
Cross-attention	Bottleneck	-0.047	[-0.070, -0.026]	1.00	< 0.001
ECG-only	ResNet1D	+0.118	[+0.089, +0.148]	0.052	< 0.001
ResNet1D	Cross-attention	-0.060	[-0.087, -0.035]	0.95	< 0.001
TCN	Cross-attention	-0.052	[-0.080, -0.023]	1.00	< 0.001
Gradient boosting (v1)	Cross-attention	-0.099	[-0.133, -0.066]	0.049	< 0.001

Only the comparison against gradient boosting satisfies the rule that both tests must agree after correction; the others are described as suggestive or not different.

B.2 Perturbation, noise and transfer experiments

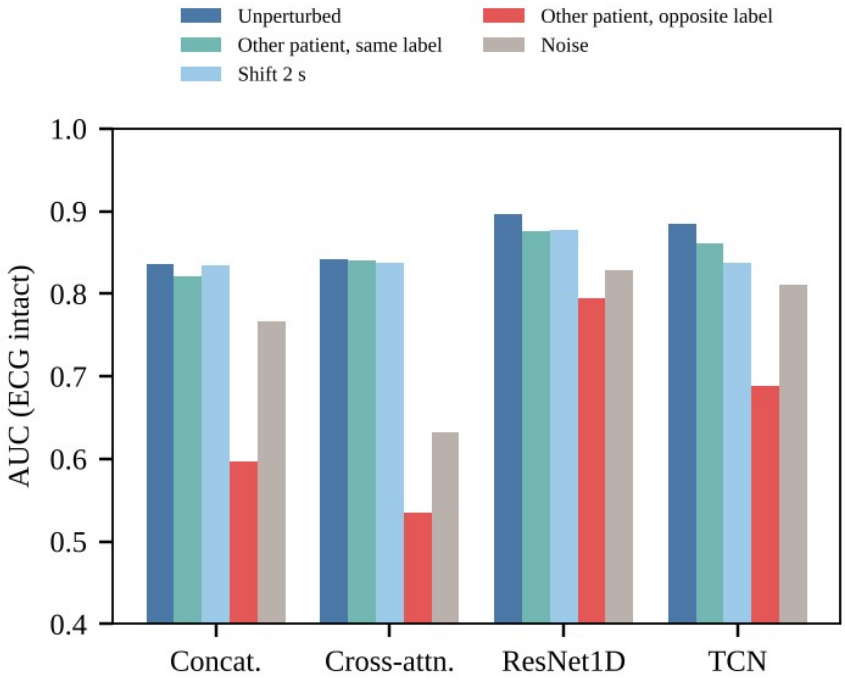


Figure 11: Test AUC when the PPG is perturbed and the ECG is intact (seed-0 checkpoints, five test folds).

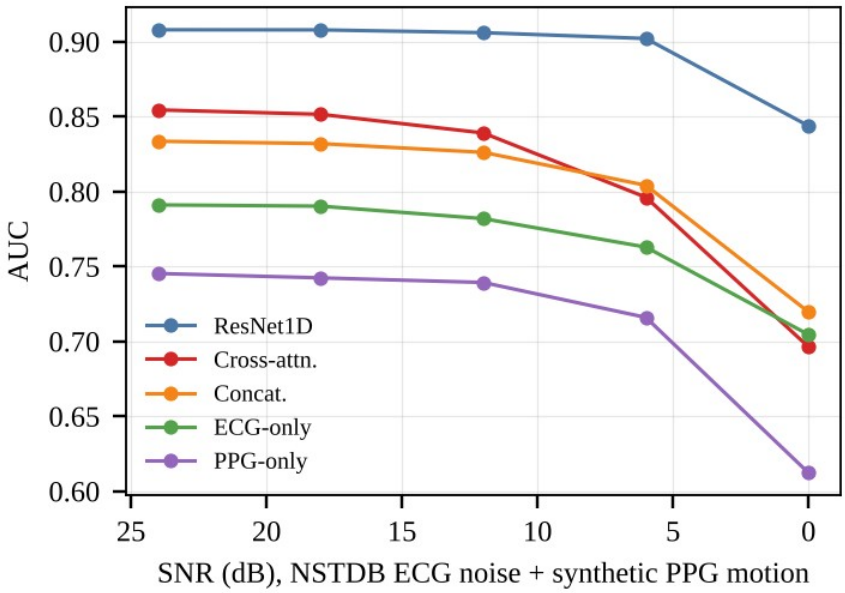


Figure 12: AUC versus injected noise level, mean over noise types (real Noise Stress Test noise on the ECG, synthetic motion artefact on the PPG).

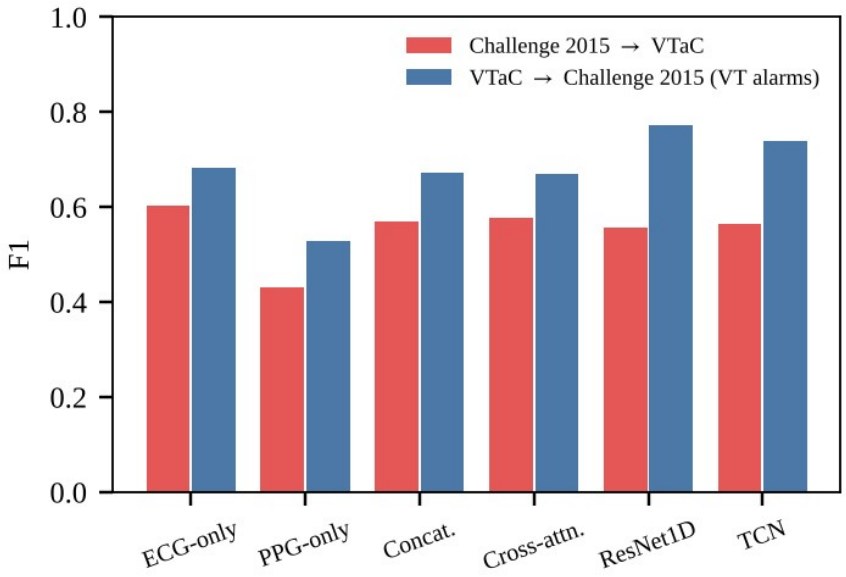


Figure 13: Cross-dataset transfer measured by F1.

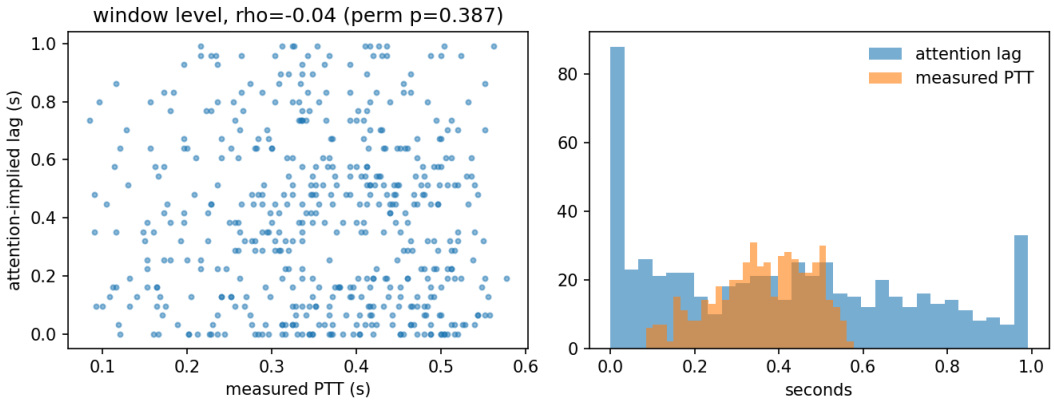


Figure 14: Attention-implied lag versus measured pulse transit time (left) and the two lag distributions (right).

B.3 Other results

AF screening (MIMIC PERform AF, 35 subjects, subject-wise cross-validation, low power). A gradient-boosting model on RR-irregularity features reached subject-level AUC 1.00, significantly better than the best neural model at 0.93. Fusion beat a PPG-only network by +0.28 subject AUC, and distilling a PPG-only student from the fusion teacher gave no benefit.

Foundation model. A frozen PaPaGei linear probe reached F1 0.58 on Challenge 2015, and the best PaPaGei fusion model was about 0.04 F1 below scratch cross-attention [11].

Label reliability. In VTaC, 7.7% of events have split annotator votes, and the models do much worse on them: cross-attention scores 0.704 accuracy there against 0.854 on unanimous events.

Safety layer (concatenation model only). With a 10% miss tolerance at 90% confidence, a threshold could be certified in 42 of 50 folds. In those folds the layer suppressed 22.5% of false alarms and missed 2.9% of true alarms, and calibration

lowered the expected calibration error from 0.167 to 0.121 (temperature scaling [30], exact Clopper-Pearson bound [31]).

Annexure C: Implementation Evidence

We did not build an end-user application. The project includes a static results dashboard (site/index.html) and a replay page (site/replay.html) that display the stored leaderboards. The screenshot below shows the dashboard header and the Challenge 2015 leaderboard. One caveat: the dashboard labels the best VTaC model "attn CNN-RNN + features", but the recipe code only uses the hand-crafted features for the "_feat" architectures, and the winning architecture is the plain attention CNN-GRU without them, which is also consistent with its parameter count (258,857).

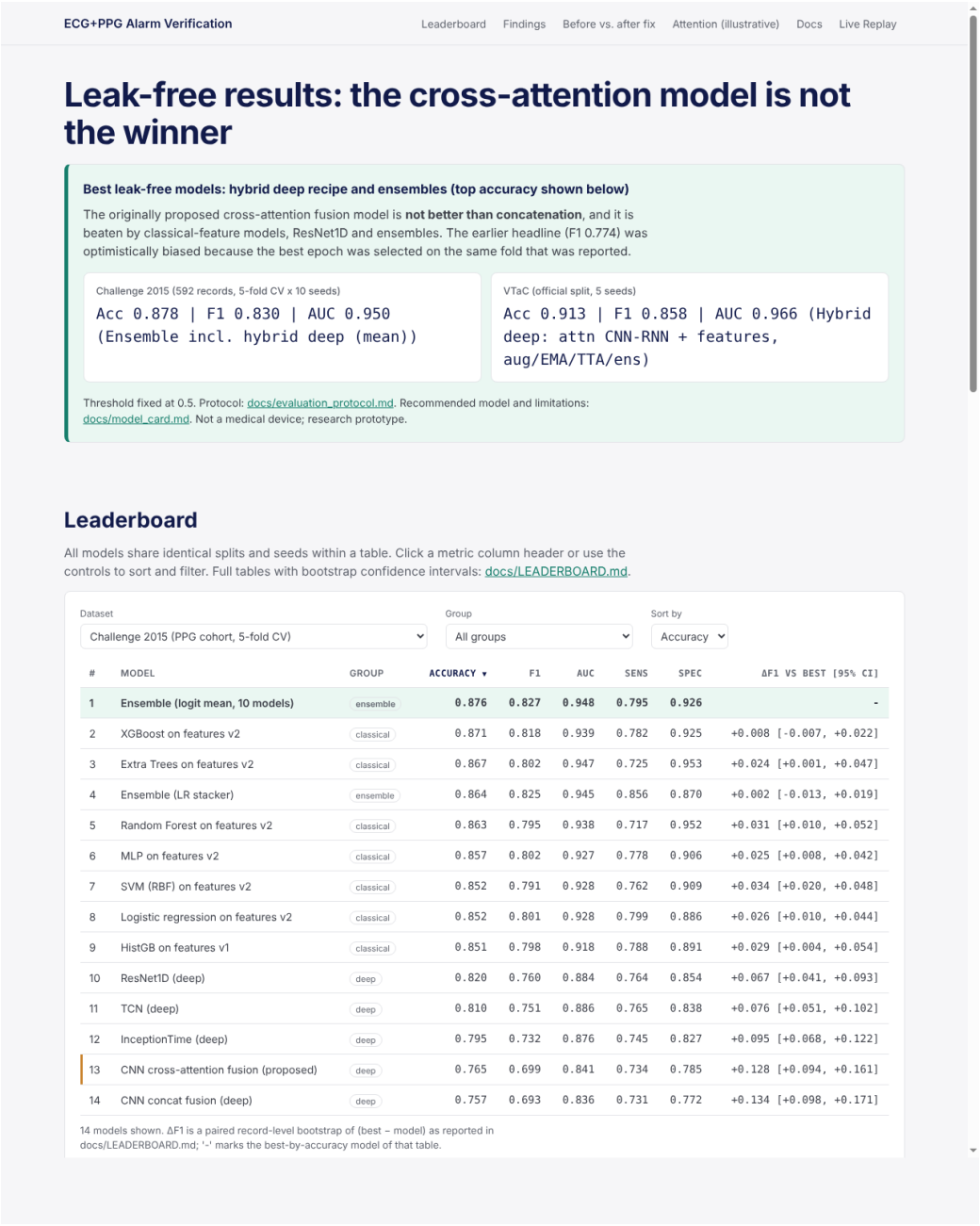


Figure 15: Screenshot of the results dashboard (site/index.html).

Annexure D: Team Contributions

The contribution of each member is to be written by the team before submission; it has been left blank here because it cannot be derived from the repository.

Table 11: Team contributions.

Member	**Contribution (to be completed by the team)**
Veerbhadra Mahant	
Ved Jadhav	
Sharvil Patil	